MIL-M-45000D
30 July 1986
SUPERSEDING
MIL-M-45000C
8 February 1985

MILITARY SPECIFICATION

MAINTENANCE FIXTURE, RADIATOR, AIR-OPERATED, HYDRAULIC

This specification is approved for use by all departments and agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers air-operated hydraulic maintenance fixtures, designed for use in cleaning, testing and repairing all types of automotive radiators.

* 1.2 Classification. Fixtures shall be of the following types as specified (see 6.2):

Type I - 350 pound capacity
Type II - 1000 pound capacity

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies of ASTM Publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.1).

3.2 Design. The radiator maintenance fixture shall be designed to facilitate the cleaning, testing and repairing of all types of automotive radiators within the size and weight range addressed in this specification. The fixture shall primarily consist of a base; air-operated hydraulic cylinder assembly; lift control; overhanging arm; and a radiator holding assembly. The raising and lowering of the radiator holding assembly shall be accomplished hydraulically by an air-operated lift control valve. An overhanging arm shall be securely attached to the top of the piston shaft and shall be capable of being manually rotated in a horizontal plane of 360 degrees regardless of what position the piston shaft is located in within its traveling range. This shall enable the fixture to raise a radiator from one tank, swing it to another tank and lower it to the desired depth within the traveling range. The radiator shall be held to the radiator holding assembly by manual clamps or other similar device(s). The holding assembly shall be designed to allow the radiator to be rotated both vertically and horizontally while securely clamped to the assembly. The holding assembly shall have a locking mechanism so that the radiator can be securely locked in any desired position within its rotation range, horizontally and vertically. This shall facilitate the exact positioning of the radiator for convenience of the required cleaning, testing and repairing operations. The radiator maintenance fixture shall be of the manufacturer's current commercial design. All parts subject to wear or replacement shall be accessible for adjustment and repair.

3.2.1 Controls. All operating controls shall be located convenient to the operator's normal work station.

3.2.2 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the fixture that present safety hazards. The safety devices shall not interfere with operation of the fixture. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair. All fixture parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.3).

3.3 Construction.

3.3.1 Interchangeability. All parts and components required in the fixture assembly shall be manufactured to definite dimensions and tolerances to provide interchangeability of replacement parts and to assure that any part replaced shall not adversely affect the function or performance of the original fixture in any manner whatsoever.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

3.3.2 Castings and forgings. All castings shall be free from blow holes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatches, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.6 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting.

3.3.7 Welding, brazing and soldering. Welding, brazing and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.3.8 Materials. Materials not specifically designated shall be of the quality required to assure that the radiator maintenance fixture can meet all of the requirements set forth in this specification. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity and intent of the materials originally selected and specified. It is not permissible to use reclaimed parts as is, or rebuild from scrap piles or other used equipment.

3.3.9 Lubrication. Means shall be provided to ensure that all moving, rotating and sliding parts are supplied with adequate lubrication. All oil holes, grease fittings and filler caps shall be accessible. The hydraulic oil must be of a quality to meet the maintenance fixture manufacturer's requirements and all of the operating conditions set forth herein.

3.3.10 Safety devices. A safety feature shall be provided to prevent the over hanging lifting arm and holding assembly from dropping suddenly due to failure of pneumatic or hydraulic supply or connections thereto.

3.4 Detail of components. The radiator maintenance fixture shall consist primarily of a base; an air-operated hydraulic cylinder assembly; lift control; overhanging arm; and a radiator holding assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

3.4.1 Base. The base shall be of steel or cast iron construction and shall have not less than four (4) bolt holes, of sufficient diameter, to securely anchor the unit to the floor.

3.4.2 Cylinder assembly. The cylinder shall be made of hydraulic tubing or seamless honed steel tubing and shall be joined to the base plate by welding, or other means, that will render the assembly leakproof at 200 pounds per square inch gage (psig) air pressure.

3.4.2.1 Piston. The outside surface of the piston shall function as a bearing and a sealer within the cylinder assembly. The piston shall be capable of rotating not less than 180 degrees within the cylinder. Means shall be provided to prevent vertical over-travel of the piston. The piston shall be hydraulically activated, by air, and shall be capable of safely lifting, stopping and lowering the overhanging arm (see 3.4.4), with attached radiator holding assembly (see 3.4.5), while lifting the full load capacity specified herein (see 3.4.5.2). The piston shall have a vertical travel range to meet the requirements herein (see 3.4.5.3).

3.4.2.2 Bearing surfaces. Metal bearing surfaces shall be provided to enable the piston to be raised and lowered in a smooth motion without sticking or chattering. Spacing of the bearings shall be as needed to prevent the possibility of binding the piston under any of the operating conditions set forth herein.

3.4.2.3 Packing glands. Packing glands shall be constructed from an oil and water resistant material, and shall be fitted to provide an effective seal against loss of air pressure or oil. A means of adjustment shall be provided to compensate for wear.

3.4.2.4 Wiper ring. A wiper ring, located at the top of the cylinder, shall be provided to prevent the entry of dirt and water into the cylinder.

3.4.3 Lift control. Vertical travel of the piston shall be governed by a 3-position air control valve (up, neutral, and down). The valve shall be a self-closing type, located within reach of the operator. The valve shall be shielded or positioned to prevent accidental contact of the control lever by the operator's body. The neutral position of the control lever shall automatically stop the motion of the lifting piston at any desired location within the piston's traveling range. When the control lever is in the neutral position the piston shall remain stable and hold the load capacity without creeping in either direction (see 3.5.3).

* 3.4.4 Overhanging arm. An overhanging arm, capable of being manually rotated 360 degrees in a horizontal plane, shall be attached to the top of the lifting piston shaft. The arm shall be capable of full rotation throughout the vertical traveling range of the piston (see 3.4.5.1). The arm shall carry the radiator holding assembly (see 3.4.5). The length of the arm shall be sufficient to provide 6 inches clearance between the housing of the cylinder assembly and the radiator holding assembly. The arm shall be capable of supporting the radiator holding assembly with the full capacity load without cracking, breaking, or permanent deformation.

3.4.5 Radiator holding assembly. The radiator holding assembly shall be rigidly suspended from the free end of the overhanging arm. The assembly shall hold the radiators in various positions for the necessary cleaning, testing and repairing operations. The radiators shall be held to the assembly by a manually operated clamp or similar device. The assembly shall be of metal construction, however, all metal parts that come in actual contact with the radiator shall be cushioned with a suitable material that will prevent damage to the radiator due to the pressure exerted by the

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:11Z
Check the source to verify that this is the current version before use.

clamping or holding method used. The holding assembly shall be capable of securely holding a radiator of the maximum size and weight specified herein (see 3.4.5.2).

3.4.5.1 Universal adjustment. The holding assembly shall be universally adjustable to the extent that it shall provide not less than 340 degree rotation on both the vertical and horizontal planes. The joints providing the rotation shall be capable of being locked in any desired position within the rotation range, without slippage, while holding the load capacity specified herein (see 3.4.5.2).

* 3.4.5.2 Capacity. The holding assembly shall be adjustable to accomodate various sizes of radiators. The type I holding assembly shall accomodate radiators up to and including 45 inches in height or width with core thicknesses ranging from 1 inch up to and including 6 inches. The type II holding assembly shall accomodate radiators up to and including 57 inches in height or width with core thickness ranging from 1 inch up to and including 16 inches. Type I assemblies shall safely handle radiators weighing 350 pounds and less and type II assemblies shall safely handle radiators weighing 1000 pounds and less.

* 3.4.5.3 Vertical travel. Vertical travel of the holding assembly (with the aid of the piston travel) shall be sufficient to allow an operator to raise the lowest point of a radiator from floor level, to a height not less than 45 inches from the floor for the type I and 57 inches from the floor for the type II, swing it over the edge of a test tank, and submerge it to any desired depth within the traveling range.

3.5 Performance characteristics.

3.5.1 Air pressure. The radiator maintenance fixture shall operate satisfactorily on compressed air at any pressure within the range of 100 to 150 psig.

* 3.5.2 Temporary overload. The radiator maintenance fixture shall be capable of withstanding 10 minutes of operation under the following conditions. The radiator maintenance fixture shall support the temporary overload without cracking, breaking or permanent deformation.

a. 200 psig air pressure (plus or minus 5 psig).

b. 525 pound load (plus or minus 5 pounds) for a type I fixture or 1500 pound load (plus or minus 15 pounds) for a type II fixture.

* 3.5.3 Anti-creep. When loaded to its rated capacity, 350 pounds for type I or 1000 pounds for type II, and raised to any point in the upper half of the holding assembly travel range, the maintenance fixture shall be capable of standing for a period of 30 minutes with a measured creep of not more than 1 inch in either direction.

3.6 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely affixed to each fixture. The nameplate shall contain the information listed below. If the fixture is a special model, the nameplate shall contain the model number of the basic standard fixture with a prefix "S" after the number.

Nomenclature
Manufacturer's Name
Manufacturer's Model Designation
Manufacturer's Serial Number
Power Input Characteristics and Ratings

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

Date of Manufacture
Contract Number or Order Number
National Stock Number

* 3.6.1 Lubrication chart or plate. A corrosion-resistant metallic nameplate, with permanently legible characters, written in the English language, shall be attached to each fixture in a location conveniently visible to the operator. If a chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear plastic, permanently sealed sheets with suitable means for mounting. The following information shall be furnished on the chart and plate:

Points of application
Service interval
Type of lubrication
Viscosity
Military or Federal specification number of lubricant

3.7 Workmanship. Workmanship of the radiator maintenance fixture and accessories shall be commensurate with the requirements specified herein and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be applied to the preproduction model or initial production items (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1. Failure of the item to pass the examination or test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

* 4.4.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material either indicated or actual, nor does it commit the government to acceptance of defective material.

* 4.5 Examination. Visually and manually examine the fixture to determine conformance with requirements of 3.2 through 3.4.3, 3.4.5, and 3.6 through 3.7.1. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examinations may be applied to the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the fixtures and their components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure to provide objective evidence that all parts are manufactured to definite standards, clearances and tolerances so that all replacement parts will not degrade the form, fit or function of the end item (see 3.3.1), shall be cause for rejection.

4.6 Tests.

4.6.1 Operational test. The radiator maintenance fixture shall be connected to a source of compressed air at a constant pressure of 150 psig (plus or minus 5 psig). The fixture shall be operated at no load for a minimum of 30 minutes during which time the controls, hydraulic system, piston, and all other operating components of the fixture shall be actuated and observed to determine proper operation.

* 4.6.2 Functional test. An automotive radiator weighing 350 pounds (minimum), and having minimum height, width, and thickness dimensions of 45 X 45 X 6 inches respectively or a test specimen having the same dimensions and weight for the type I fixture or an automotive radiator weighing 1000 pounds (minimum) and having minimum height, width, and thickness dimensions of 57 X 57 X 16 inches respectively, or a test specimen of the same dimensions and weight, shall be clamped securely in the radiator holding fixture. The fixture shall then be tested for compliance with the following:

(a) The piston shall not bind in the cylinder or chatter during the raising and lowering motion of the holding assembly (see 3.4.2.2).
(b) Accessibility of lift control from normal operating position, its function, and if properly positioned or guarded (see 3.4.3).
(c) Length of overhanging arm and the clearance between the cylinder assembly housing and the radiator holding assembly (see 3.4.4).
(d) Degree of rotation of overhanging arm in the horizontal plane throughout the full travel range of the piston (see 3.4.4).
(e) Degree of rotation of holding assembly (see 3.4.5.1).
(f) Vertical travel range of holding assembly (see 3.4.5.3).

Source: https://assist.dla.mil - Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

* 4.6.3 Overload test. The maintenance fixture shall be connected to a source of compressed air at a constant pressure of 200 psig (plus or minus 5 psig). A radiator, or test specimen, weighing 525 pounds (plus or minus 5 pounds) for a type I fixture or 1500 pounds (plus or minus 15 pounds) for a type II fixture, shall be clamped securely in the radiator holding assembly. The fixture shall then be operated throughout the vertical travel range of the holding assembly for a period of 10 minutes. At the end of the test, the fixture shall be carefully examined for damage or permanent deformation to the cylinder assembly, overhanging arm, or radiator holding assembly. Checks for oil leaks shall also be made (see 3.4.2.3 and 3.5.2).

4.6.4 Anti-creep test. The maintenance fixture shall be connected to a source of compressed air at a constant pressure within 100 to 150 psig. The radiator or test specimen used for this test shall be within the weight and dimensional capacity of the fixture. The radiator or test specimen shall be clamped securely in the radiator holding assembly and raised a distance of 23 inches from the floor. The distance from the floor to a reference mark on the radiator or test specimen shall be carefully measured and recorded, and the unit allowed to stand for a period of 30 minutes. At the end of the 30 minute period, the distance from the floor to the reference mark shall be remeasured and recorded. Any creep in either direction from the original distance shall not exceed one inch (see 3.5.3).

* 4.6.5 Performance test. The maintenance fixture shall be set up under the conditions specified in 4.6.4 with the addition of a radiator test tank or test tank mock-up. The upper edge of the type I test tank shall be 38 to 42 inches from the floor. The upper edge of the type II test tank shall be 50 to 55 inches from the floor. The radiator shall be raised to the required height, swung over the edge of the tank, and lowered to the point that would be necessary for complete submergence of the radiator if the tank contained water. The assembly shall then be raised, swung back over the edge of the tank, and positioned at five different angles approximately 85 degrees apart, for a total angular rotation of 340 degrees in the vertical plane and 340 degrees in the horizontal plane. At the end of the test, the holding assembly shall be lowered and the radiator removed. The fixture shall then be carefully examined for damage (see 3.4.5, 3.4.5.1, and 3.4.5.3).

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or Industrial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. All components of the radiator maintenance fixture shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Unit protection. The unit method of protection for the maintenance fixture shall conform to method I of MIL-P-116. Immediately after cleaning and drying, apply a coat of type P-11 preservative of MIL-P-116 to all exterior ferrous metal surfaces of the maintenance fixture which have not been painted or plated. Wrap the preserved surfaces with barrier material conforming MIL-B-121, grade A. Secure barrier with tape conforming to PPP-T-60, type IV. Place the piston in the innermost position and secure to prevent movement. Technical publications shall be packaged in accordance with method IC-1 of MIL-P-116 and attached to the basic unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

5.1.2 Industrial. The maintenance fixture shall be preserved in accordance with applicable provisions of ASTM D 3951-82.

5.2 Packing. Packing shall be level A, B, or Industrial (see 6.2.1).

5.2.1 Level A. The radiator maintenance fixture shall be packed in a shipping container conforming to PPP-B-621, class 2. Blocking, bracing, cushioning, and anchoring shall conform to the requirements of MIL-STD-1186. The container shall be closed and strapped in accordance with the appendix of the box specification.

5.2.2 Level B. Packing requirements shall be the same as specified for level A, except the shipping container shall conform to class 1 in accordance with PPP-B-621.

5.2.3 Industrial. The maintenance fixture shall be packed in accordance with applicable provisions of ASTM D 3951-82.

5.3 Marking.

5.3.1 Military. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of ASTM D3951-82.

6 NOTES

6.1 Intended use. The radiator maintenance fixture covered by this specification is intended for use in reconditioning automotive radiators. The fixture is used to raise radiators over the edge of a tank and submerge them for cleaning, testing and repairing operations. It is also used for holding radiators at any desired height and position necessary for the performance of the maintenance operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article inspection, when required (see 3.1).
(c) Painting, if different (see 3.3.6).
(d) Nameplate, if different (see 3.6).
(e) First article inspection, if different (see 4.3).
(f) Selection of preservation level (see 5.1).
(g) Selection of packing level (see 5.2).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Subject term (key word) listing.

AMSC number
Acquisition requirements
Air pressure
Anti-creep
Anti-creep test
Base
Blaring surface .
Capacity
Castings and forgings
Classification
Classification of inspections
Cleaning and drying
Construction
Contract data requirements
Cylinder assembly
Design
Detail of components
Distribution statement
Examination
Fastening devices
First article inspection
Functional test
Government documents
Handbooks
Intended use
Interchangeability
Lift control
Lubrication
Lubrication chart or plate
Maintenance fixture, radiator, air-operated, hydraulic
Marking
Materials
Military specification
Nameplate
Operational test
Ordering data
Overhanging arm
Overload test
Packaging
Packaging inspection
Packing
Packing glands
Painting
Performance characteristics
Performance test
Piston
Preservation
Quality Assurance Provisions

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:11Z
Check the source to verify that this is the current version before use.

Quality conformance inspection
Radiator holding assembly
Responsibility for compliance
Responsibility for inspection
Safety and Health Requirements
Safety devices
Scope
Specifications
Standards
Subject term listings
Surface
Temporary overload
Tests
Threads
Unit protection
Universal adjustment
Vertical travel
Welding, brazing and soldering
Wiper ring
Workmanship

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, and deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AL
Air Force - 99

Review activities:
Army -
Navy -
Air Force - 82

Preparing activity:
Army - AL

Project No. 4910-0614

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research and
Development Engineering Center
ATTN: SMCAR-EST-S/C. Overman
Rock Island, IL 61299-7300

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:11Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:11Z
Check the source to verify that this is the current version before use.